Pseudocode Practice Exercises for Code Champs 12-Feb-2026

🟢 Exercise 1: Secure Exam Entry System

You are building a system for an examination hall.

The program must:

1. Ask the student for:
    - AGE
    - HAS_ID (TRUE or FALSE)
    - HAS_ADMIT_CARD (TRUE or FALSE)
2. The student is allowed to enter ONLY IF:
    - AGE is GREATER THAN OR EQUAL TO 16
    - AND HAS_ID is TRUE
    - AND HAS_ADMIT_CARD is TRUE
3. If all conditions are satisfied:
    - DISPLAY "Entry Allowed"
4. If any condition fails:
    - DISPLAY "Entry Denied"
5. Additionally:
    - If AGE is LESS THAN 0, DISPLAY "Invalid Age" and EXIT the program immediately.

Your program must correctly combine logical operators.

🟡 Exercise 2: Smart Billing Machine

You are creating a billing machine for a store.

The program must:

1. Ask the user for:
    - TOTAL_AMOUNT
    - IS_MEMBER (TRUE or FALSE)
2. Apply discounts using these rules:
    - If TOTAL_AMOUNT is GREATER THAN OR EQUAL TO 200
      → Give 25% discount

    - Else if TOTAL_AMOUNT is GREATER THAN OR EQUAL TO 100
      → Give 10% discount
    - Otherwise
      → No discount
3. After discount:
    - If IS_MEMBER is TRUE
      → Give an additional 5% discount on the remaining amount
4. Finally:
    - DISPLAY the final payable amount.

The logic must apply discounts in the correct order.

🔵 Exercise 3: Controlled Number Analyzer

The program must:

1. Ask the user how many numbers they want to enter.
2. Repeat that many times.
3. For each number entered:
    - If number is negative → skip it using CONTINUE
    - If number equals 100 → immediately stop the loop using BREAK
    - If number is divisible by 5 → count it
4. After the loop ends:
    - DISPLAY how many valid numbers were divisible by 5.

This tests:

- Loop control
- MOD
- CONTINUE
- BREAK

🟣 Exercise 4: ATM Withdrawal with Validation

Initial account balance = 500

The program must:

1. Ask the user how much they want to withdraw.

2. If the amount is:
    - Less than or equal to 0 → DISPLAY "Invalid amount" and EXIT
    - Greater than balance → DISPLAY "Insufficient Funds"
    - Otherwise → subtract from balance and DISPLAY remaining balance
3. After a successful withdrawal:
    - If balance becomes 0 → DISPLAY "Account Empty"

This problem checks correct IF ordering.

🔴 Exercise 5: Prime-Like Checker

Ask the user to enter a number greater than 1.

Your program must:

1. Check whether the number has any divisors between 2 and (number - 1).
2. If any divisor is found:
    - DISPLAY "Not Prime"
    - Stop checking further (use BREAK)
3. If no divisors are found:
    - DISPLAY "Prime Number"

Important:

- If number is LESS THAN OR EQUAL TO 1 → DISPLAY "Invalid Input" and EXIT.

This requires:

- Loop
- MOD
- Logical thinking

🟠 Exercise 6: Grade Report Generator

Ask the student to enter:

- Number of subjects

Then:

1. Repeat for each subject:
    - READ score

    - Add score to total
    - If score LESS THAN 40 → count it as a failed subject
2. After loop:
    - Calculate average
    - DISPLAY average
    - DISPLAY number of failed subjects
3. Final result:
    - If failed subjects GREATER THAN 0
      → DISPLAY "Overall Result: Fail"
    - Otherwise
      → DISPLAY "Overall Result: Pass"

● Exercise 7: Number Pattern Controller

Ask the user for a number N.

Print numbers from 1 to N.

However:

- If a number is divisible by 3 → DISPLAY "Three"
- If divisible by 7 → DISPLAY "Seven"
- If divisible by BOTH → DISPLAY "ThreeSeven"
- Otherwise → display the number

Order of conditions is important.

● Exercise 8: Login With System Shutdown

Correct password = "secure123"

The system must:

1. Allow maximum 4 attempts.
2. If password is correct:
    - DISPLAY "Login Successful"
    - STOP the program
3. If wrong:

    - DISPLAY "Wrong Password"
4. If user types "shutdown":
    - EXIT immediately
5. If 4 wrong attempts happen:
    - DISPLAY "System Locked"

This tests:

- REPEAT UNTIL or REPEAT FROM
- BREAK / EXIT
- String comparison
- Attempt counter

🟡 **Exercise 9: Number Reversal System**

Ask user for a positive number.

Your program must:

1. Reverse the digits.
2. Example:
    - 543 → 345
    - 120 → 21

If number is negative:

- DISPLAY "Invalid Input"
- EXIT

You must use:

- MOD
- Division
- Loop

🔵 **Exercise 10: Mini Voting Booth**

Ask the user how many voters are voting.

For each voter:

- READ AGE

- READ IS_CITIZEN (TRUE or FALSE)

Rules:

- If AGE ≥ 18 AND IS_CITIZEN = TRUE → count as valid vote
- Otherwise → count as invalid vote

After all voters:

- DISPLAY total valid votes
- DISPLAY total invalid votes